

Introduction to the Git

Supervised by :FOUAD Imane

Devlopped by :OUAJIH Houda

4- Initializing a Git repository

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ouaji>git --version
git version 2.55.0.windows.5

C:\Users\ouaji>git config --global user.name "Houda Ouajih"

C:\Users\ouaji>git config --global user.email Houda.OUAJIH@um6p.ma

C:\Users\ouaji>mkdir tpgit

C:\Users\ouaji>cd tpgit

C:\Users\ouaji\tpgit>git init
Initialized empty Git repository in C:/Users/ouaji/tpgit/.git/
```

```
C:\Users\ouaji\tpgit>dir /a
Volume in drive C has no label.
Volume Serial Number is 76AA-AB9D

Directory of C:\Users\ouaji\tpgit

09/17/2026  11:11 AM    <DIR>          .
09/17/2026  11:06 AM    <DIR>          ..
09/17/2026  11:11 AM    <DIR>          .git
               0 File(s)              0 bytes
               3 Dir(s)  166,747,705,344 bytes free

C:\Users\ouaji\tpgit>type nul >fruits.txt
```

```

C:\Users\ouaji\tpgit>echo banana >>fruits.txt

C:\Users\ouaji\tpgit>echo avocado >>fruits.txt

C:\Users\ouaji\tpgit>git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    fruits.txt

nothing added to commit but untracked files present (use "git add" to track)

C:\Users\ouaji\tpgit>git add
Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config set advice.addEmptyPathsSpec false"

C:\Users\ouaji\tpgit>git add ./fruits.txt

C:\Users\ouaji\tpgit>git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   fruits.txt

C:\Users\ouaji\tpgit>git commit -m "[adding 2 fruits to fruits.txt banana and avocado]"
[master (root-commit) 45d19ae] [adding 2 fruits to fruits.txt banana and avocado]
1 file changed, 2 insertions(+)
create mode 100644 fruits.txt

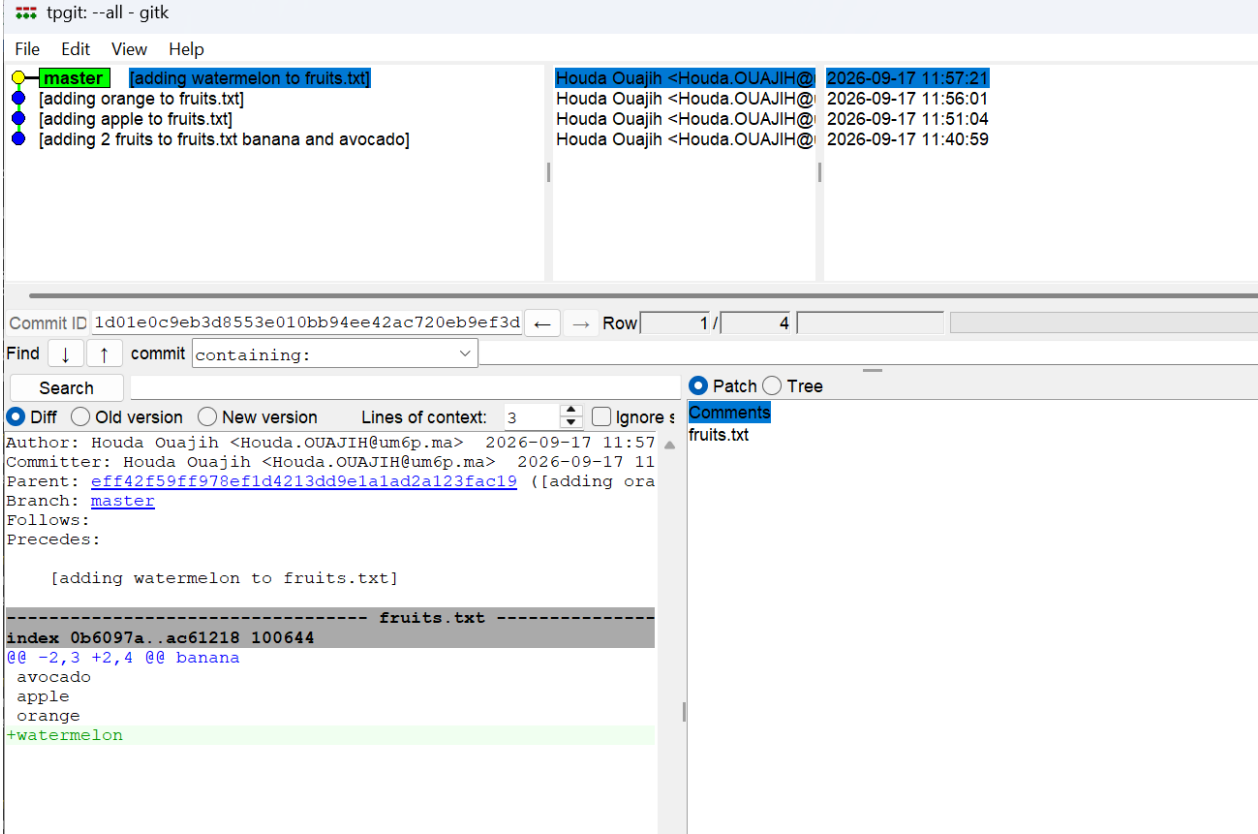
C:\Users\ouaji\tpgit>git status
On branch master
nothing to commit, working tree clean

C:\Users\ouaji\tpgit>git log
commit 45d19aecdfbf1f9ce211e12fcb31d6104435458c (HEAD -> master)
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 11:40:59 2026 +0100

    [adding 2 fruits to fruits.txt banana and avocado]

```

The same for adding apple,orange and watermelon



5- Git Branches

Vegetables branch

```
C:\Users\ouaji\tpgit>git branch vegetables

C:\Users\ouaji\tpgit>git checkout vegetables
Switched to branch 'vegetables'

C:\Users\ouaji\tpgit>

C:\Users\ouaji\tpgit>git branch
  master
* vegetables

C:\Users\ouaji\tpgit>type nul >vegetables.txt

C:\Users\ouaji\tpgit>git add ./vegetables.txt

C:\Users\ouaji\tpgit>git commit -m "[adding vegetables.txt]"
[vegetables 604e30c] [adding vegetables.txt]
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 vegetables.txt
```

```
C:\Users\ouaji\tpgit>echo potato >>vegetables.txt

C:\Users\ouaji\tpgit>git add ./vegetables.txt

C:\Users\ouaji\tpgit>git commit -m "[adding potato to vegetables.txt]"
[vegetables d57b56b] [adding potato to vegetables.txt]
1 file changed, 4 insertions(+)

C:\Users\ouaji\tpgit>echo pepper >>vegetables.txt

C:\Users\ouaji\tpgit>git add ./vegetables.txt

C:\Users\ouaji\tpgit>git commit -m "[adding pepper to vegetables.txt]"
[vegetables 53a9ac5] [adding pepper to vegetables.txt]
1 file changed, 1 insertion(+)

C:\Users\ouaji\tpgit>echo pumpkin >>vegetables.txt

C:\Users\ouaji\tpgit>git add ./vegetables.txt

C:\Users\ouaji\tpgit>git commit -m "[adding pumpkin to vegetables.txt]"
[vegetables bc1c1cf] [adding pumpkin to vegetables.txt]
1 file changed, 1 insertion(+)

C:\Users\ouaji\tpgit>git log
commit bc1c1cf9a715cale08f8439016da577717f75ec (HEAD -> vegetables)
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 12:27:14 2026 +0100

    [adding pumpkin to vegetables.txt]

commit 53a9ac549b74fe4825590ce0dd34f42ea8082c0a
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 12:26:45 2026 +0100

    [adding pepper to vegetables.txt]

commit d57b56b66d1a77ddd6510f21cea5155f51fd170
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 12:26:00 2026 +0100

    [adding potato to vegetables.txt]

commit 604e30c57877d3ccf59130e192740cecf5015211
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 12:09:08 2026 +0100

    [adding vegetables.txt]
```

The screenshot shows the Git GUI interface. At the top, the command line displays 'tpgit: --all - gitk'. Below this, a list of commits is shown, including 'vegetables' (adding pumpkin to vegetables.txt) and 'master' (adding watermelon to fruits.txt). The right pane shows a diff view for the 'vegetables.txt' file, highlighting the addition of 'pumpkin' to the list of fruits. The bottom pane shows the commit details for the selected commit, including the author, committer, parent, branch, and the diff itself.

Sauces branch

```
PS C:\Users\ouaji\tpgit> git branch sauces
PS C:\Users\ouaji\tpgit> git checkout sauces
Switched to branch 'sauces'
```

```
C:\Users\ouaji\tpgit>type nul>sauces.txt
C:\Users\ouaji\tpgit>echo ketchup>>sauces.txt
C:\Users\ouaji\tpgit>add ./sauces.txt
'add' is not recognized as an internal or external command,
operable program or batch file.
C:\Users\ouaji\tpgit>git add sauces.txt
C:\Users\ouaji\tpgit>git commit -m "[adding ketchup to sauces.txt]"
[sauces ald5174] [adding ketchup to sauces.txt]
1 file changed, 1 insertion(+)
create mode 100644 sauces.txt
C:\Users\ouaji\tpgit>echo barbecue>>sauces.txt
C:\Users\ouaji\tpgit>git add sauces.txt
C:\Users\ouaji\tpgit>git commit -m "[adding barbecue to sauces.txt]"
[sauces ea4a49e] [adding barbecue to sauces.txt]
1 file changed, 1 insertion(+)
C:\Users\ouaji\tpgit>git branch
  master
* sauces
  vegetables
C:\Users\ouaji\tpgit>git log
commit ea4a49e910cc3cfed769abc438ce13dceb226d8f (HEAD -> sauces)
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 19:41:06 2026 +0100

    [adding barbecue to sauces.txt]

commit ald5174731513f69041fdb7b84dd26016c419499
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
Date: Thu Sep 17 12:43:12 2026 +0100

    [adding ketchup to sauces.txt]

commit bc1c1cfd9a715ca1e08f8439016da577717f75ec (vegetables)
Author: Houda Ouajih <Houda.OUAJIH@um6p.ma>
```

● sauces	[adding barbecue to sauces.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding ketchup to sauces.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
● vegetables	[adding pumpkin to vegetables.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding pepper to vegetables.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding potato to vegetables.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding vegetables.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
● master	[adding watermelon to fruits.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding orange to fruits.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding apple to fruits.txt]	Houda Ouajih <Houda.OUAJIH@um6p.ma>
●	[adding 2 fruits to fruits.txt banana and avocado]	Houda Ouajih <Houda.OUAJIH@um6p.ma>

6- Git Merges

```
C:\Users\ouaji>cd tpgit

C:\Users\ouaji\tpgit>git checkout vegetables
Switched to branch 'vegetables'

C:\Users\ouaji\tpgit>git status
On branch vegetables
nothing to commit, working tree clean

C:\Users\ouaji\tpgit>git merge master
Already up to date.

C:\Users\ouaji\tpgit>git checkout master
Switched to branch 'master'

C:\Users\ouaji\tpgit>git merge vegetables
Updating 1d01e0c..bcl1c1cf
Fast-forward
 vegetables.txt | 6 ++++++
 1 file changed, 6 insertions(+)
 create mode 100644 vegetables.txt
```

```
C:\Users\ouaji\tpgit>git checkout sauces
Switched to branch 'sauces'
```

```
C:\Users\ouaji\tpgit>git merge master
Already up to date.
```

```
C:\Users\ouaji\tpgit>git checkout master
Switched to branch 'master'
```

```
C:\Users\ouaji\tpgit>git merge sauces
Updating bcl1c1cf..ea4a49e
Fast-forward
 sauces.txt | 2 ++
 1 file changed, 2 insertions(+)
 create mode 100644 sauces.txt
```

```
C:\Users\ouaji\tpgit>git log --graph --oneline --decorate --all
* ea4a49e (HEAD -> sauces, vegetables, master) [adding barbecue to sauces.txt]
* a1d5174 [adding ketchup to sauces.txt]
* bcl1c1cf [adding pumpkin to vegetables.txt]
* 53a9ac5 [adding pepper to vegetables.txt]
* d57b56b [adding potato to vegetables.txt]
* 604e30c [adding vegetables.txt]
* 1d01e0c [adding watermelon to fruits.txt]
* eff42f5 [adding orange to fruits.txt]
* 06da61d [adding apple to fruits.txt]
* 45d19ae [adding 2 fruits to fruits.txt banana and avocado]
```

7-Remote repository

```
C:\Users\ouaji\tpgit>git remote add origin https://github.com/Houda-Ouajih/tpgit.git
C:\Users\ouaji\tpgit>git remote -v
origin https://github.com/Houda-Ouajih/tpgit.git (fetch)
origin https://github.com/Houda-Ouajih/tpgit.git (push)

C:\Users\ouaji\tpgit>git push -u origin master
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote: https://github.com/Houda-Ouajih/tpgit/pull/new/master
remote:
To https://github.com/Houda-Ouajih/tpgit.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

 master ▾  1 Branch  0 Tags  Add file ▾  Code ▾

 Houda-Ouajih [adding barbecue to sauces.txt] ea4a49e · 1 hour ago  10 Commits

 fruits.txt [adding watermelon to fruits.txt] 9 hours ago

 sauces.txt [adding barbecue to sauces.txt] 1 hour ago

 vegetables.txt [adding pumpkin to vegetables.txt] 8 hours ago

8-Collaborative Work

```
C:\Users\ouaji\tp-git-binome>git clone https://github.com/malak8677/tpgit.git tp-git-binome
Cloning into 'tp-git-binome'...
remote: Enumerating objects: 27, done.
remote: Counting objects: 100% (27/27), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 27 (delta 1), reused 27 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (27/27), done.
Resolving deltas: 100% (1/1), done.
```

```
C:\Users\ouaji\tp-git-binome>log --graph --oneline --decorate --all
'log' is not recognized as an internal or external command,
operable program or batch file.
```

```
C:\Users\ouaji\tp-git-binome>git log --graph --oneline --decorate --all
* b645652 (HEAD -> master, origin/master, origin/HEAD) Add barbecue sauce to sauces.txt
* 4dd4788 Add sauces.txt with Tomato Sauce
* 65ae04b Add spinach to vegetables.txt
* 7180a7e add potato to vegetables.txt
* 72df67b add carrot to vegetables.txt
* 707bbeb Add strawberry and kiwi to fruits list
* 990dc02 Add pineapple and mango to fruits list
* d83b335 Add banana and grape to fruits list
* 295f982 Add apple and orange to fruits list
```

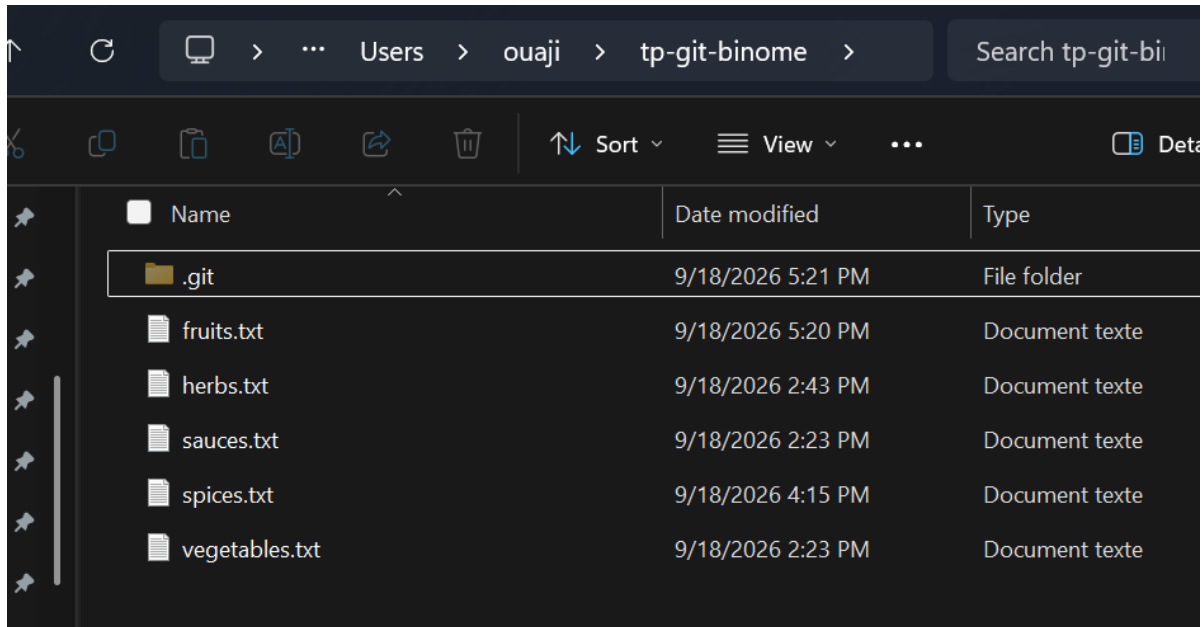
```
C:\Users\ouaji\tp-git-binome>git checkout master
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(use "git push" to publish your local commits)

C:\Users\ouaji\tp-git-binome>git merge herbs
Already up to date.

C:\Users\ouaji\tp-git-binome>git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 263 bytes | 11.00 KiB/s, done.
From https://github.com/malak8677/tpgit
   b645652..89a9e12  master    -> origin/master
Merge made by the 'ort' strategy.
 README.md | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
```

```
C:\Users\ouaji\tp-git-binome>git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 774 bytes | 387.00 KiB/s, done.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/malak8677/tpgit.git
   89a9e12..e2094fa  master -> master
```

9-MERGE WITHOUT CONFLITS



10-MERGE WITH CONFLITS

```
C:\Users\ouaji\tp-git-binome>git add ./fruits.txt

C:\Users\ouaji\tp-git-binome>git commit -m "[change with watermelon]"
[master 5c69d5e] [change with watermelon]
1 file changed, 1 insertion(+), 1 deletion(-)

C:\Users\ouaji\tp-git-binome>git push
To https://github.com/malak8677/tpgit.git
! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/malak8677/tpgit.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

C:\Users\ouaji\tp-git-binome>git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 319 bytes | 10.00 KiB/s, done.
From https://github.com/malak8677/tpgit
 1b2cfde..91affc7  master    -> origin/master
Auto-merging fruits.txt
CONFLICT (content): Merge conflict in fruits.txt
Automatic merge failed; fix conflicts and then commit the result.
```

```
C:\Users\ouaji\tp-git-binome>git status
On branch master
Your branch and 'origin/master' have diverged,
and have 1 and 1 different commits each, respectively.
(use "git pull" if you want to integrate the remote branch with yours)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Changes to be committed:
  deleted:    REDAME.md

Unmerged paths:
  (use "git add <file>..." to mark resolution)
  both modified: fruits.txt
```

Conflicts problem solving

```
C:\Users\ouaji\tp-git-binome>notepad fruits.txt

C:\Users\ouaji\tp-git-binome>git add ./fruits.txt

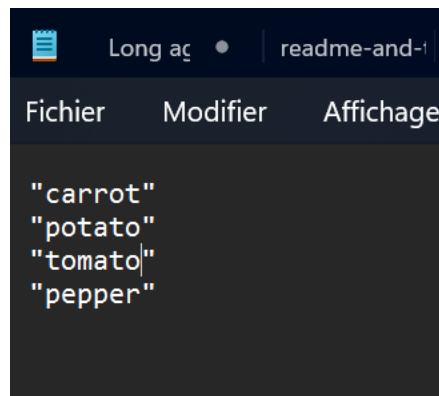
C:\Users\ouaji\tp-git-binome>git commit -m "[conflits fixed]"
[master 4a0d9e2] [conflits fixed]

C:\Users\ouaji\tp-git-binome>git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 678 bytes | 339.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/malak8677/tpgit.git
 91affc7..4a0d9e2  master -> master
```

with deleting >> and === and others change to reorganize it
we changed the line apple , the first line, with watermelon for me , and kaki
for my pair

```
watermelon
kaki
orange
banana
grape
pineapple
mango
strawberry
kiwi
```

Reverse conflicts also solved



```
Long aç • readme-and-t
Fichier Modifier Affichage

"carrot"
"potato"
"tomato"
"pepper"
```

11-Documentation

```
C:\Users\ouaji\tp-git-binome>notepad README.md

C:\Users\ouaji\tp-git-binome>git add README.md

C:\Users\ouaji\tp-git-binome>git commit -m "[file README.md added]"
[master 7754875] [file README.md added]
1 file changed, 50 insertions(+)
create mode 100644 README.md

C:\Users\ouaji\tp-git-binome>git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 863 bytes | 863.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/malak8677/tpgit.git
326165b..7754875 master -> master
```